

Panjam - Reimagining Music Learning to Support Healthy Aging and Wellbeing

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ABSTRACT

In this paper, we present Panjam - a prototype VR application that reimagines music learning for seniors to support healthy aging and wellbeing. Panjam uses the intuitive interface of the steelpan paired with immersion, presence, multi-sensory feedback, and gamification of the VR system to induce active engagement, self-driven practice and learning.

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ABSTRACT

1 INTRODUCTION

Scientific research ([Global Council on Brain Health 2020]) suggests that music can have a profound impact on brain health and mental wellbeing. It is well established that listening to music can improve the lives of seniors but playing a musical instrument is equivalent to a full body workout for the brain. As the world population ages rapidly, learning how to apply more of music's potential to support the brain and healthy aging holds the promise of improving the lives of many seniors.

2 PROBLEM STATEMENT

Learning to play a musical instrument can be particularly challenging for seniors ([Dascálua et al. 2014]) - requiring motivation, practice and at times physical discomfort. These challenges limit the adoption of musical instrument among seniors limiting the potential positive impact music could have on their lives.

3 METHODOLOGY

Through surveys, semi-structured interviews and secondary research we have identified four focal pain points most seniors experience learning to play: (1) Beginners Blues/Boredom (2) Physical Constraints (3) Mobility and (4) Tuition. Each of these pain points

are mapped to a core set of features that propose a solution to address that pain point. These features are components of the overall system design.

4 REFRAMING PAIN POINTS INTO DESIGN COUNTERPARTS

To address **beginner's blues**, we developed a real time multi-sensory system that uses haptic, audio and visual feedback that makes the experience for users more engaging, immediate and responsive. To address **physical constraints**, we use the intuitive interface of the steelpan ([Rossing 2000]) that is easy to learn and enjoyable to play. To address **mobility**, we transport seniors out of their everyday lives and immerse them in virtual environments where they feel stimulated and inspired. To address **tuition**, we promote a learn by playing method that uses gamification to reward achievements ([Chou 2016]), allowing seniors to learn at their own pace.

5 SYSTEM DESIGN

5.1 Virtual Environments

A user can choose among various virtual environments to explore, giving them a sense of presence and mobility. Some environments are responsive to the user's actions while others determine the acoustics.

5.2 The Virtual Instrument

The virtual steelpan is programmed to produce midi outputs. The value of the output for each note will be determined by the game mechanics ([Hunicke et al. 2004]) (the rules of the system) and play dynamics (what the user does). Midi messages - note on/off, pitch, velocity will match the acoustics of the instrument. The velocity specifies how quickly or forcefully the note is struck.

5.3 Input controls

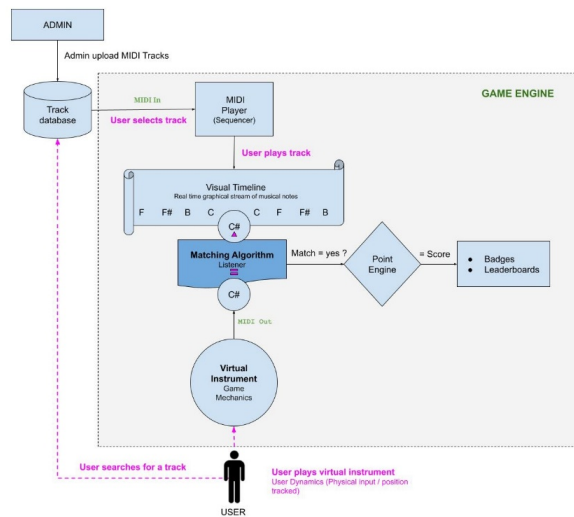
Hand controllers transform into virtual pan sticks, which when struck against the virtual steelpan will resonate a rich buoyancy of tones synonymous with the real instrument.

5.4 Feedback systems

The system generates auditory, haptic, and visual feedback creating a multi-sensory (audio-spatial) experience for the user. Audio and visuals are designed in tandem, using visuals to reinforce physical interaction and scaffold learning.

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6 GAME MECHANICS



System design schematic

6.1 Database

A user can select a preloaded midi track from a database of music composed to be played by the virtual instrument. Once a track is selected to initiate game play the user can select play to start.

6.2 Visual timeline

Each track is composed of an arrangement of notes mapped to a specific timeline. The timeline is visually represented to the user in 3D space as a stream of dynamic “running” notes highlighting the note currently being played in real time.

6.3 Game modes

6.3.1 Level one-Fundamentals. A note will be highlighted as a visual cue. Once struck by the user the next note on the timeline will be highlighted. This will be repeated until all the notes in the track are played.

6.3.2 Level two-Memory and Repetition. An initial sequence of notes will be highlighted. The user must remember the sequence of notes to play. Once the notes in the sequence are played correctly, an additional note will be added to the sequence. This will be repeated until all the notes in the track are played.

6.3.3 Level three-Dexterity and Precision. A note will be highlighted and a user must play it before the next note on the timeline advances. The user will need to play each note at the right time to avoid missing the note.

6.3.4 Free play mode-Mastery. The user can play the instrument without any game mechanic, visual cues or score.

6.4 Point system, Time and Synchronization

To generate points, the player must strike the exact note at the precise time depending on the game mode. As the user strikes a note on the virtual instrument a matching algorithm will listen to what note the user has played and match it to the note that is current being streamed in real time. If the matching algorithm registers a match, it will then calculate a value based on the game mechanics.

7 PLAY DYNAMICS - DIMENSIONS OF EXPRESSIVE CONTROL PER NOTE

7.1 Precision (Sweet Spot)

Each note has a sweet spot and if struck precisely and at the right time will yield extra points for the user.

7.2 Hit (Strike the Pan) - Volume

Mimicking the acoustics of the real instrument the low notes (the big ones!) barely need to be tapped while the high notes (the little ones!) must be struck more forcefully.

8 CONCLUSION

Panjam has the potential to transform traditional methods of music learning and to define a whole new category. As the global population ages, the potential of music as a non-pharmacological intervention will become increasingly relevant to many millions of people. Panjam is positioned at the forefront of a new and exciting field of research where music technology, VR and cognitive performance intersects.

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